



ST0263

Tópicos Especiales en

Telemática

Escuela de Ciencias Aplicadas e
Ingeniería
2025-2

Modelling the BookStore Web Application

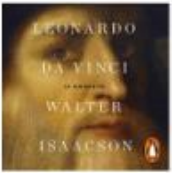
Re-Architect Approach

Description

- The application allows you to view a collection of resources, for the purposes of this scenario, “books”.
- Likewise, when the user selects one of the resources, a view is offered with detailed information about the selected resource.
- The information of the resources (books) is stored in a database.
- The application has three (views): root ("/", home), detailed description of the resources books and about.

Application Views (1)

CATALOGO DE LIBROS



Leonardo Davinci: La Biografia

\$50.000.00



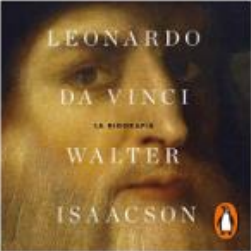
Inteligencia Genial

\$30.000.00

MI TIENDA DE LIBROS

INICIOACERCA DE

REGRESAR



LEONARDO DAVINCI:
LA BIOGRAFIA

WALTER ISAACSON

Estado: Disponible (2) eds

Precio: \$50.000.00

Autor: Walter Isaacson

Descripción: Basándose en las miles de páginas de los cuadernos manuscritos de Leonardo y nuevos descubrimientos sobre su vida y su obra, Walter Isaacson teje una narración que conecta el arte de Da Vinci con sus investigaciones científicas, y nos muestra cómo el genio del hombre más visionario de la historia nació de habilidades que todos poseemos y podemos estimular, tales como la curiosidad insaciable, la observación cuidadosa y la imaginación juguetona. Su creatividad, como la de todo gran innovador, resultó de la interacción entre la tecnología y las humanidades. Despejajó y estalló el rastro de numerosos cadáveres, dibujó los músculos que configuran al movimiento de las tablas y pintó la sonrisa más enigmática de la historia, la de la Mona Lisa. Exploró las leyes de la óptica, demostró cómo la luz incide en la córnea y logró producir esa ilusión de profundidad en la última cena.

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Application Views (2)

MI TIENDA DE LIBROS

INICIO ACERCA DE

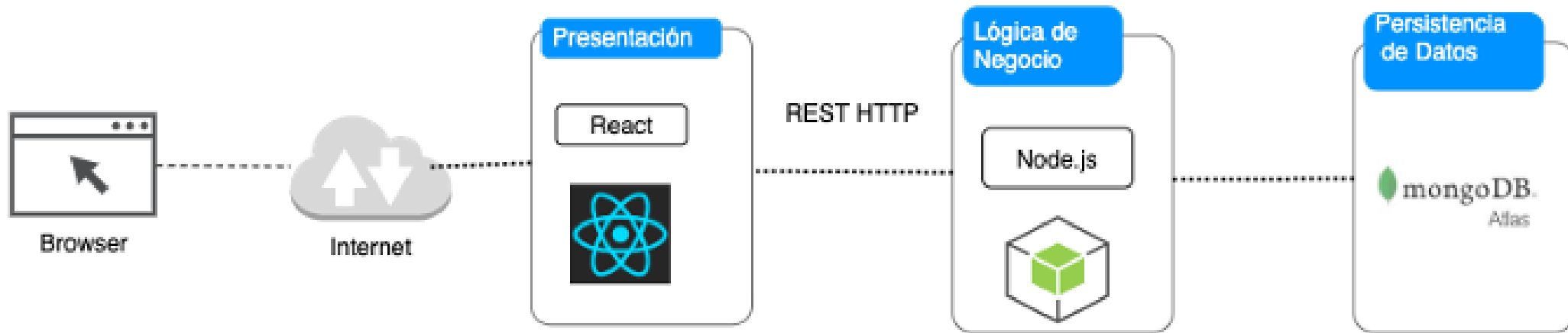
Este sitio web esta concebida como una tienda virtual de obras literarias, la cual permite obtener el listado de obras disponibles para reservar y/o comprar. El sitio fue desarrollado para que los usuarios pueda obtener información útil acerca de...

Software Architecture

Architecture Description

- It is a web application which uses a layered architectural style.
- For the purposes of this project, a three (3) layer architecture was defined: presentation layer, business logic and data persistence.
- This is an architectural style that allows each layer to run on top of its own infrastructure.
- Likewise, it is possible for a development team to focus on each layer.

High-Level Architecture



Description

- **Presentation Layer Description (Front End):**
 - This layer handles all aspects related to the user interface.
 - It is accessed through the browser and is responsible for communication with the back end, in this case using a REST API..
 - The presentation layer or front end is developed using React, which is defined as a JavaScript library for developing the presentation layer.
 - The main function of React is to render HTML code in the browser.
 - For handling styles (CSS), React-Bootstrap is used.

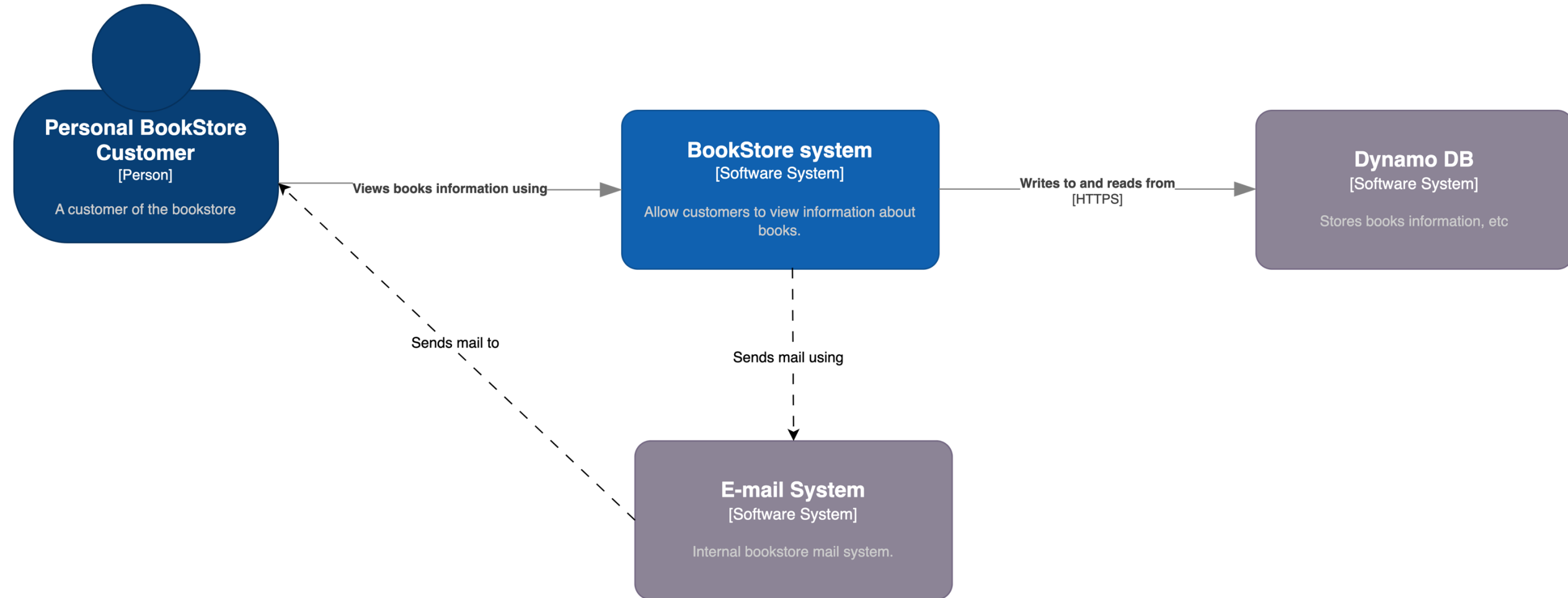
Description(2):

- Back end Description:
- In this architectural style, this is the middle layer where business logic is executed and carried out.
- The back end is developed using server-side scripting technology, in this case, Node.js.
- Additionally, we use a web application development framework called Express and Mongoose for database connection.

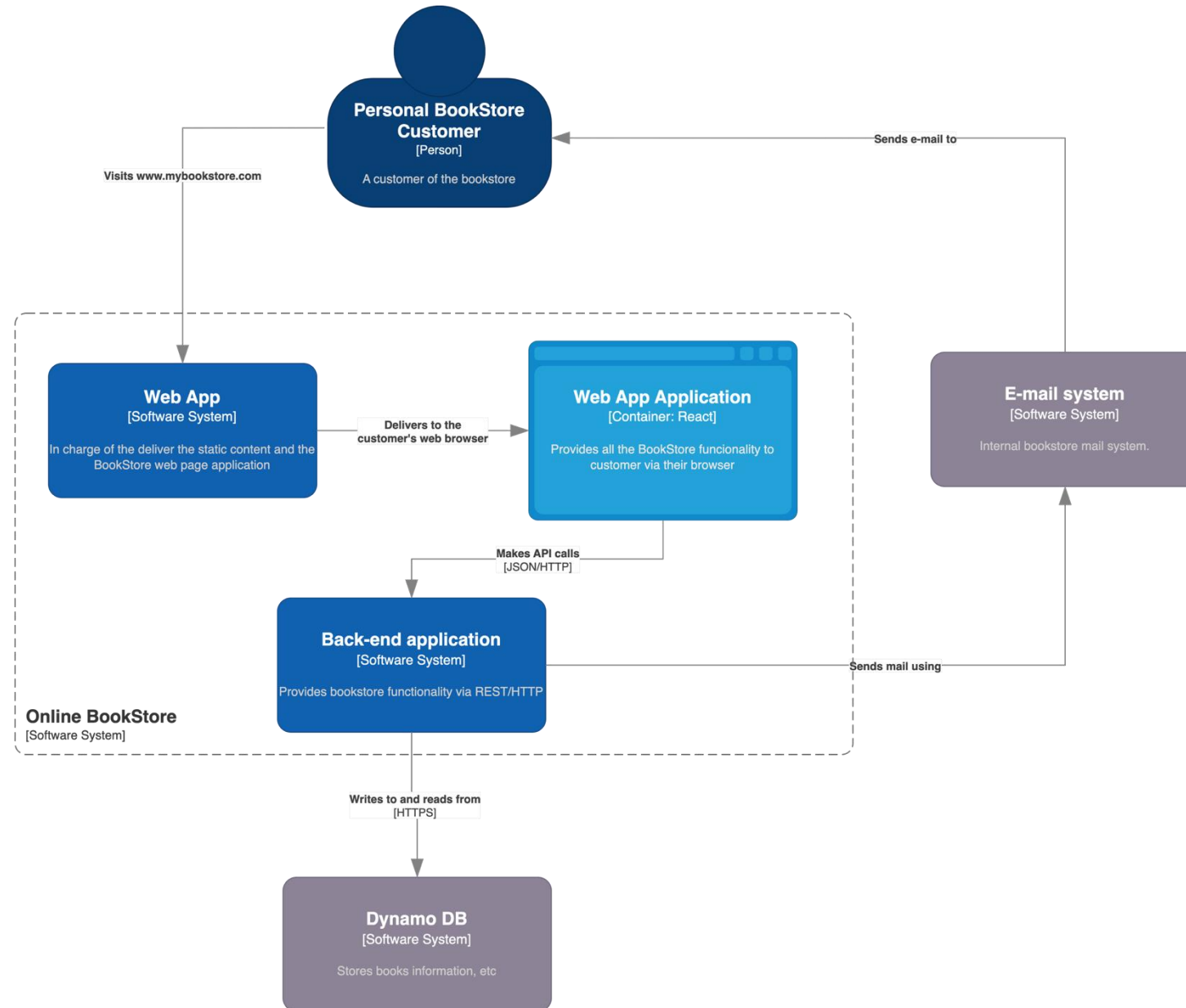
Description (3):

- Description of data persistence:
 - In this layer, the information is managed and stored.
 - In this scenario, the data is stored in a document-oriented database engine, specifically MongoDB for the purposes of this application.
 - In particular, we will be using MongoDB Atlas, which is a cloud-based NoSQL Database as a Service (DBaaS).

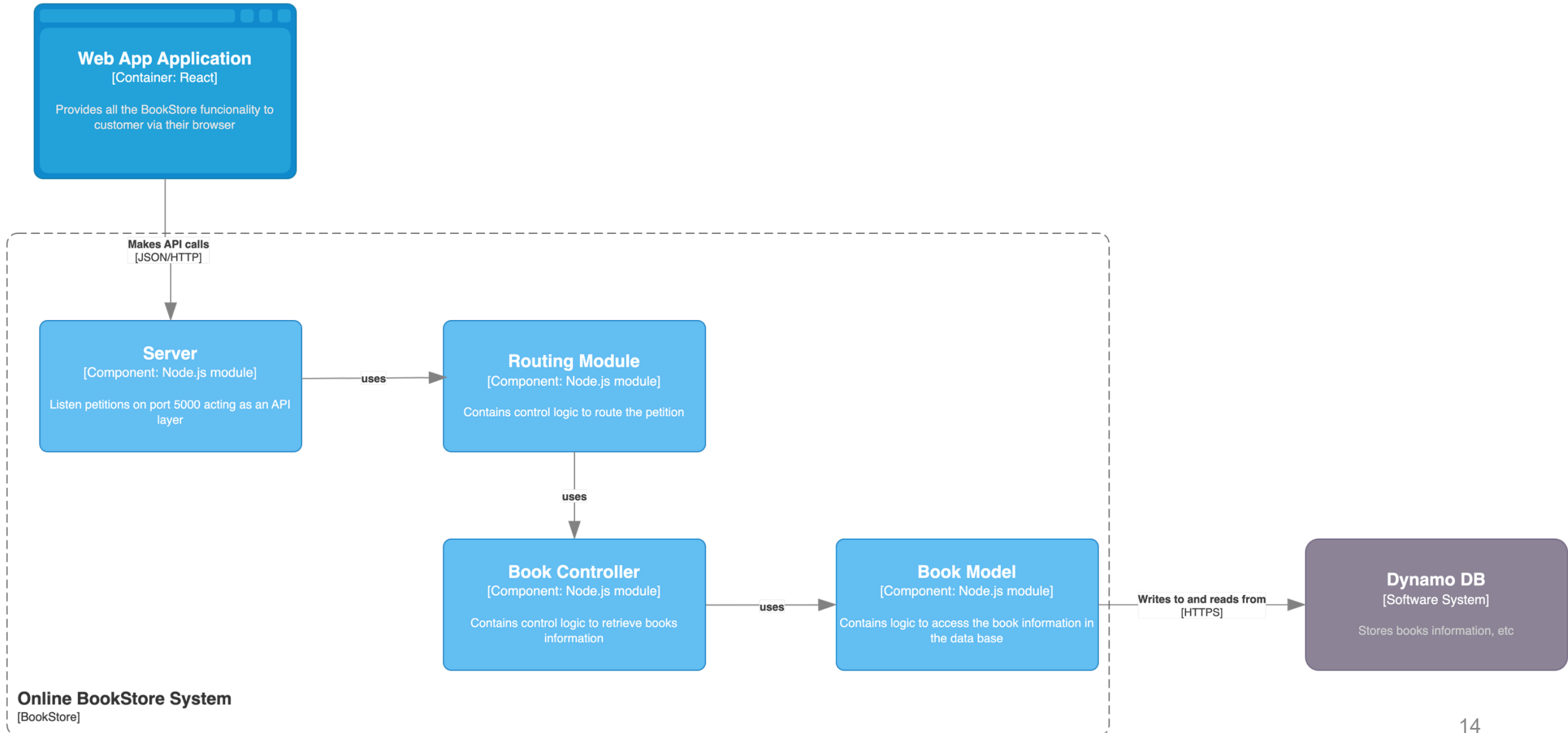
System Context Diagram



Container Diagram

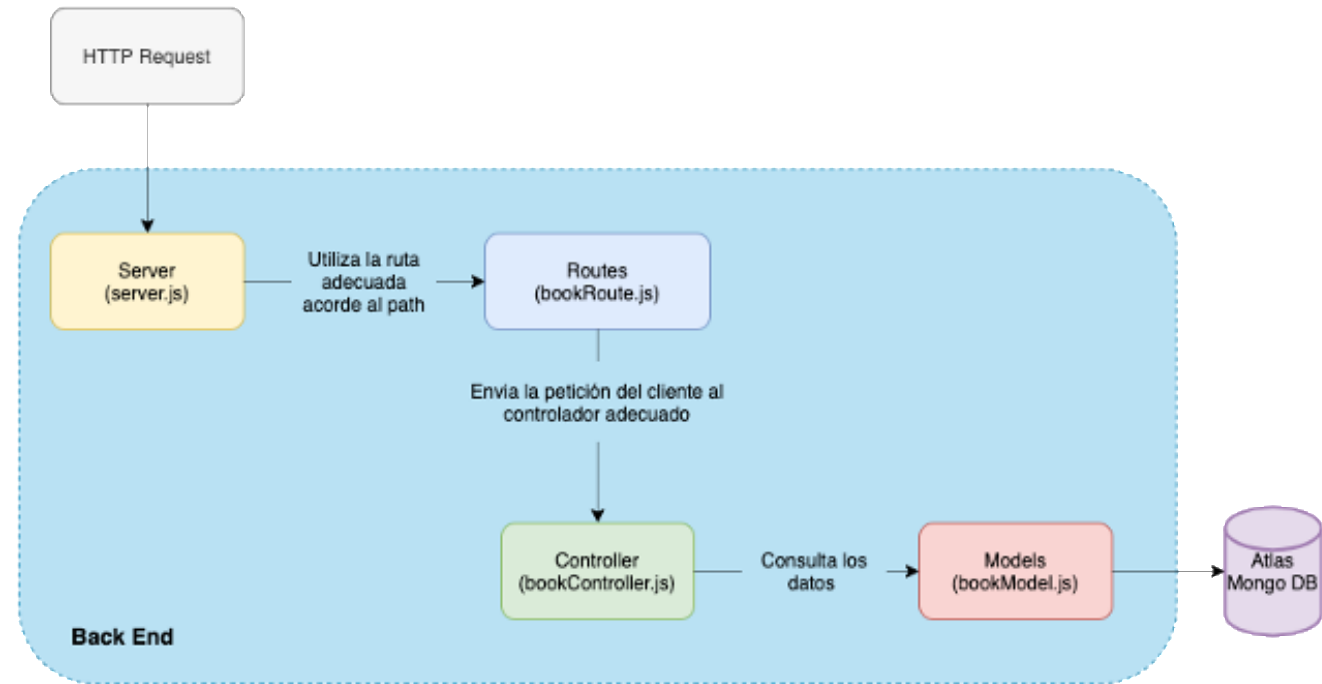


Component Diagram



Description of the Request Flow

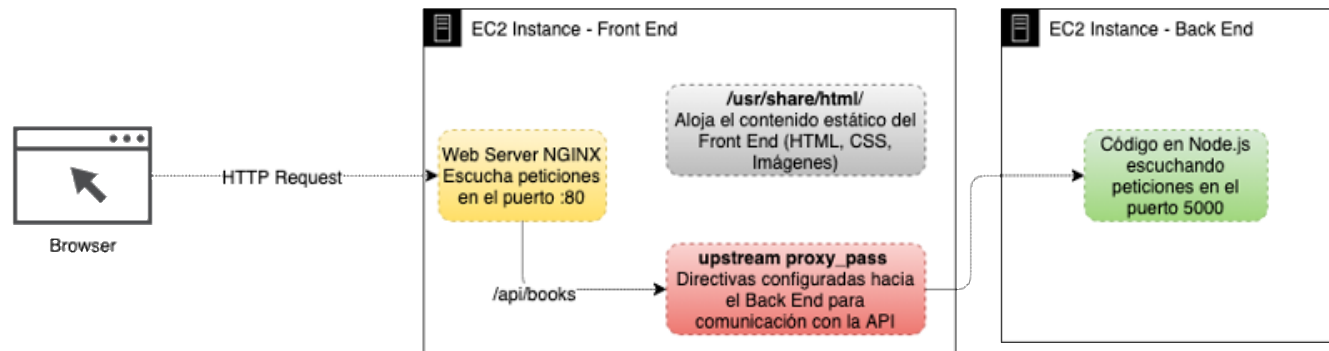
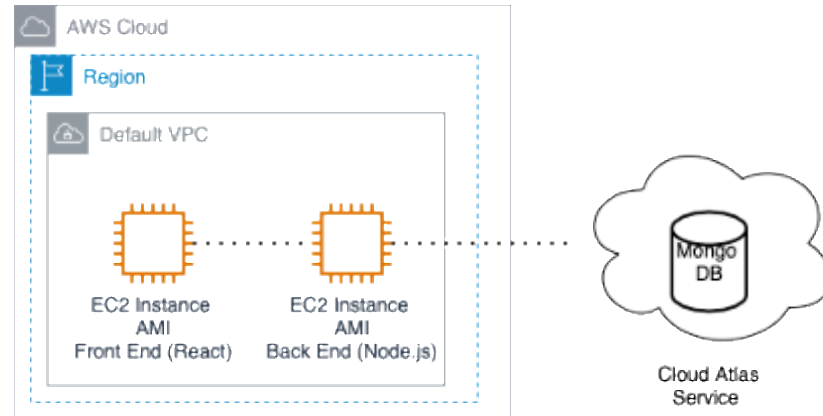
- The "server.js" module initializes the server process on port 5000 and establishes the connection with the database.
- Once it receives HTTP requests, it selects and uses the appropriate route module (bookRoute.js) for processing the incoming HTTP request.
- Here, the information is sent to the appropriate controller ("bookController") so it can fetch the requested data from the front end through the model ("bookModels.js").
- In this case, the response sent back is a JavaScript Object Notation (JSON) to the front end.
- The model is responsible for retrieving the requested data from the database.



Data

```
{
  name: 'Leonardo Davinci: La Biografia',
  image: '/images/img-ld-labiografia.jpeg',
  author: 'Walter Issacson',
  description: 'Basándose en las miles de páginas de los cuadernos manuscritos de Leonardo y nuevos descubrimientos sobre su vida y su obra, Walter Isaacson teje una narración que conecta el arte de Da Vinci con sus investigaciones científicas, y nos muestra cómo el genio del hombre más visionario de la historia nació de habilidades que todos poseemos y podemos estimular, tales como la curiosidad incansable, la observación cuidadosa y la imaginación juguetona. Su creatividad, como la de todo gran innovador, resultó de la intersección entre la tecnología y las humanidades. Despellejó y estudió el rostro de numerosos cadáveres, dibujó los músculos que configuran el movimiento de los labios y pintó la sonrisa más enigmática de la historia, la de la Mona Lisa. Exploró las leyes de la óptica, demostró como la luz incidía en la córnea y logró producir esa ilusión de profundidad en la Última cena.',
  countInStock: '2',
  price: '$50.000.00',
},
{
  name: 'Inteligencia Genial',
  image: '/images/img-ld-inteligenciagenial.jpeg',
  author: 'Michael Gelb',
  description: 'El que, para muchos, ha sido el mayor genio de todos los tiempos, Leonardo da Vinci, puede servir de inspiración a todo aquel que quiera desarrollar al máximo su potencial intelectual y su creatividad. Paso a paso, mediante ejercicios, técnicas y lecciones, este libro muestra el camino para ampliar los horizontes de la mente',
  countInStock: '3',
  price: '$30.000.00',
},
```


Actual Deployment:



Challenge

Domain

- **Catalog Service:** Manages books, authors, genres, and metadata. Provides search and filtering capabilities for users.
- **Cart Service:** Handles each user's shopping cart adding, updating, and removing items before checkout.
- **Order Service:** Processes customer orders, tracks their status (created, paid, shipped, delivered), and coordinates fulfillment.
- **Payment Service:** Integrates with payment gateways* to process transactions securely and handle confirmations or failures.
- **User Service:** Manages customer accounts, authentication, profiles, and access control.
- **Review Service:** Stores and retrieves customer reviews and ratings for books to enhance engagement.
- **Inventory Service:** Tracks stock levels across warehouses or suppliers and updates availability in real time.
- **Notification Service:** Sends transactional emails, SMS, or push notifications (order confirmations, shipment alerts, etc.).
- **Recommendation Service:** Suggests books to users based on their history, preferences, and purchases.
- **Admin / Reporting Service:** Provides dashboards and reports for managing sales, products, and performance

Asynchronous Communication

- Enable asynchronous communication and decouple microservices to improve scalability and resilience.
- Why include a broker (RabbitMQ):
 - Decoupling: Services communicate through messages, not direct calls.
 - Asynchronous processing: Long operations (payments, notifications) run in the background.
 - Reliability: Messages are stored until successfully consumed.
 - Scalability: Workload can be distributed across multiple consumers.
 - Resilience: If a service fails, messages remain in the queue and are processed once it recovers.

Challenge

- It is required to propose the following:
 - Fully scalable architecture with high availability for application deployment.
 - The application is required to be dockerized.
 - It is required to be deployed in a Kubernetes cluster using EKS
 - A minimum of 4 microservices must be developed and deployed.
 - Deliverables:
 - Date: XXX until XXXX
 - The project must be submitted in interactive virtual.
 - A video and document must be submitted.
 - The whole implementation and operation process must be explained in detail, both video and document.